Supplementary / Aegrotat Exam

Electricity and Electronics EBN111

June 2025

Assessment Information			
Maximum marks:	46	Full marks:	45
Duration of test:	90 min + (5 min before and 10 min after for OCR answer sheet completion)	Open/Closed book:	Closed
Total number of pages (this page included)		18	

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Important

1. The examination regulations of the University of Pretoria apply.
2. The departmental rules relevant to electronically graded assessments apply.
3. Answer all questions on the OCR answer sheets provided. The question number **01** in this question paper refers to row 01 of the OCR answer sheet 00/01.
4. Read the instructions on the OCR answer sheet 00/01 carefully.
5. Where applicable, answers must include units.
6. The final answers must be written as real numbers and not as fractions.
7. Use at least 4 significant figures to write numbers, including complex numbers.
8. A complex number may be written in either rectangular or polar format.
9. Exact numbers in which decimal expansions are zero can be written without the zero expansions, for example 0 V instead of 0.000 V or 3.5 A instead of 3.500 A.

INSTRUCTIONS

STEP 1 (-5 min to 0 min): Complete the front page of OCR answer sheet 01, and also fill in your student number on additional OCR answer sheets.

STEP 2 (0 min to 90 min): Do your calculations on the question paper.

STEP 3 (90 min to 100 min): Carefully copy your answers to the OCR answer sheets. Be sure to include units where applicable.

- The Assessment ID is: **2025EBN111E02**
- Use a mechanical pencil with 0.5 mm HB lead to complete the form.
- Each cell should have only one character, and the characters should not touch or cross the cell.
- **Do not use commas!** Please use full stops as decimal points.

Examples for complex numbers on OCR forms

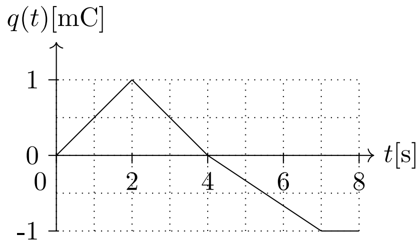
5	∠	−	9	0	°				V	
6	.	9	∠	1	5	.	9	°	μ	A
1	2	−	j	8						
j	1	5							Ω	
−	1	.	5	+	j	8	.	2	m	V

Rectangular: The j is at the start of the imaginary number.
The j is in the CENTRE of the cell.
The j does NOT cross or touch cell boundaries!

Polar: The degree sign ° is directly after the angle value.

Section 1

Questions 1 to 2 [2]	
The graph of the **charge** flowing out of the positive terminal of a battery is shown below.	

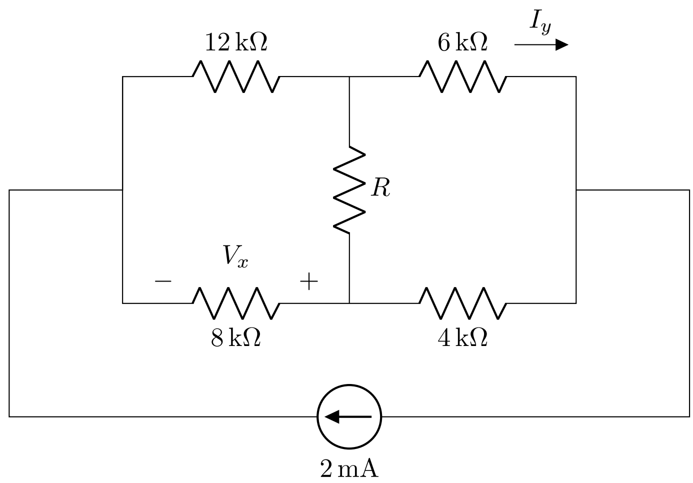


$i(t) = \frac{dq(t)}{dt}$

Looking for gradient

01	Find the current flowing at $t = 3$ s. [1] $-0.5\,mA$
02	Find the current flowing at $t = 7.5$ s. [1] $0A$

$$i(3) = \frac{0-1}{4-2}\,mA = -0.5\,mA$$

Questions 3 to 4 [4]		
		
03	Determine V_x when R =0 Ω. [2]	~~−14.4~~ V −9.6
04	Determine I_y when R is infinity. [2]	0.8 ~~A~~ mA

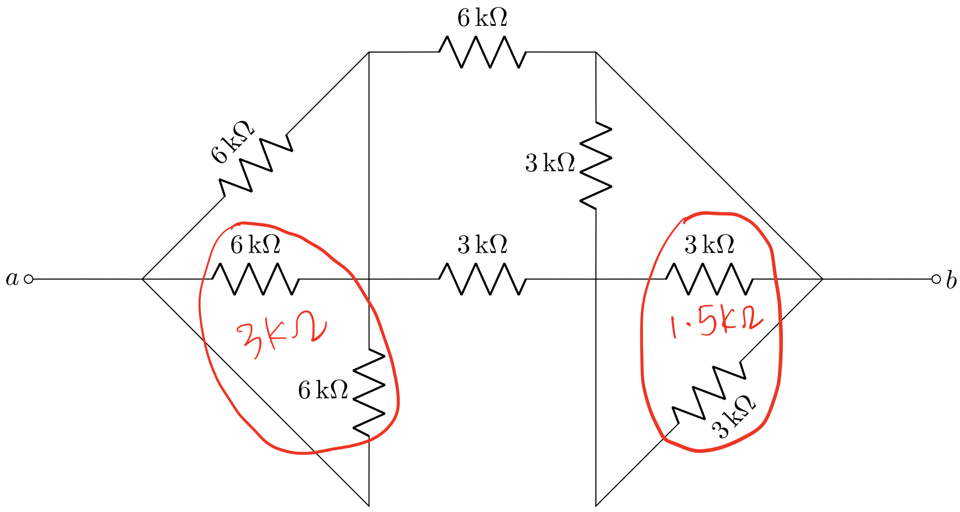
(03) $V_x = -2\,mA \times 12 // 18\ k\Omega$

$$= 2 \times \frac{12\times\cancel{1}8}{30} = \cancel{-14.4}\ V \quad -9.6$$

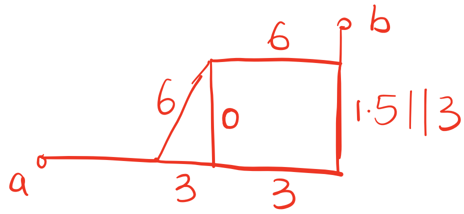
(04) R → ∞, current division.

$$I_y = 2\,mA \cdot \frac{8+4}{12+6+8+4}\ \cancel{k\Omega} = \frac{24}{30} = 0.8\cancel{A}\ mA$$

Question 5 [2]	



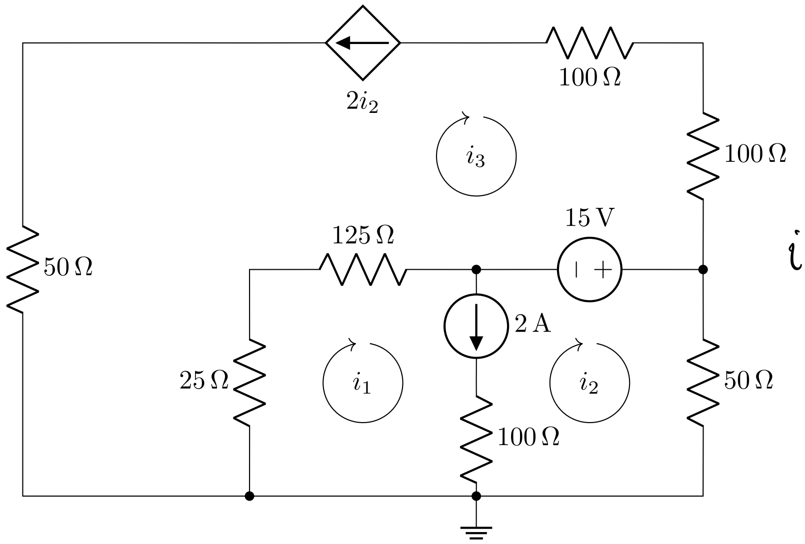
05	Calculate R_{ab} [2] 4.4kΩ



$$R_{ab} = 3 \| 6 + 6 \| (3 + 1.5 \| 3)$$

$$= 2 + 6 \| \left(3 + \frac{4.5}{4.5}\right)$$

$$= 2 + 6 \| 4 = 2 + 2.4 = 4.4\,k\Omega$$

Questions 6 to 9 [4]	
Use mesh analysis to set up the following two equations in the form: $$ai_1 + bi_2 = 2,$$ $$ci_1 + di_2 = 15.$$ Calculate the unknown parameters a, b, c, and d.	
	
06	a [1] 1
07	b [1] -1
08	c [1] 150
09	d [1] 350

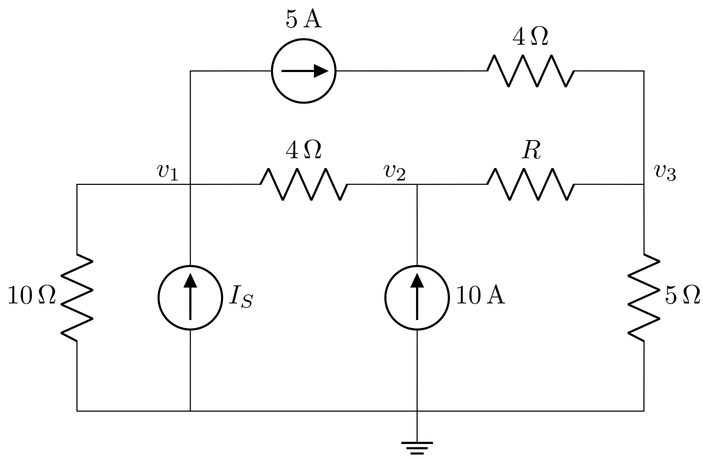
$i_1 - i_2 = 2$

$i_3 = -2i_2$

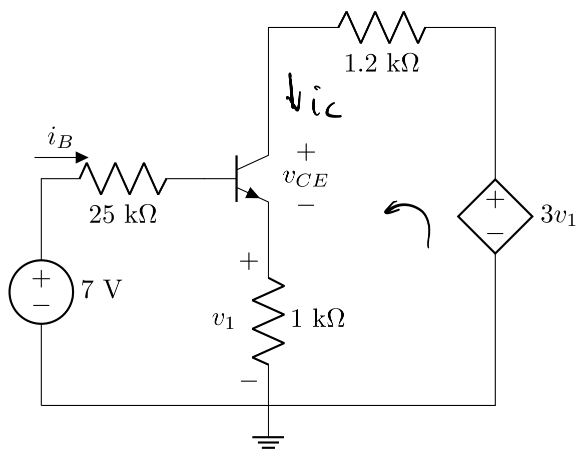
$(25+125)(i_1 - i_3) - 15 + 50i_2 = 0$

$150i_1 - 150(-2i_2) - 15 + 50i_2 = 0$

$150i_1 + 350i_2 = 15$

Questions 10 to 11 [2]		
The following matrix is constructed by inspection in terms of the nodal analysis in the circuit below. Determine R and I_S. $\begin{bmatrix} 0.35 & a & c \\ a & b & -0.5 \\ c & -0.5 & \underline{0.7} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} -2 \\ 10 \\ 5 \end{bmatrix}$		
		
10	R [1]	$2\,\Omega$
11	I_S [1]	3A

Question 12 [2]
Solve for v_{CE} if $i_B = 50\ \mu\text{A}$, $v_{BE} = 0.7$ V and $\beta = 99$.



$i_C = \beta i_B$

$V_1 = (\beta+1) i_B = 100 i_B$

12	v_{CE} [2]	4.06V

$-3V_1 + 1.2k \cdot i_C + V_{CE} + V_1 = 0$

$V_{CE} = 2V_1 - 1200\beta i_B$

$= 2 \cdot (1+\beta) i_B \cdot 1000 - 1200 \beta i_B$

$= (2000 \times 100 - 1200 \times 99) i_B$

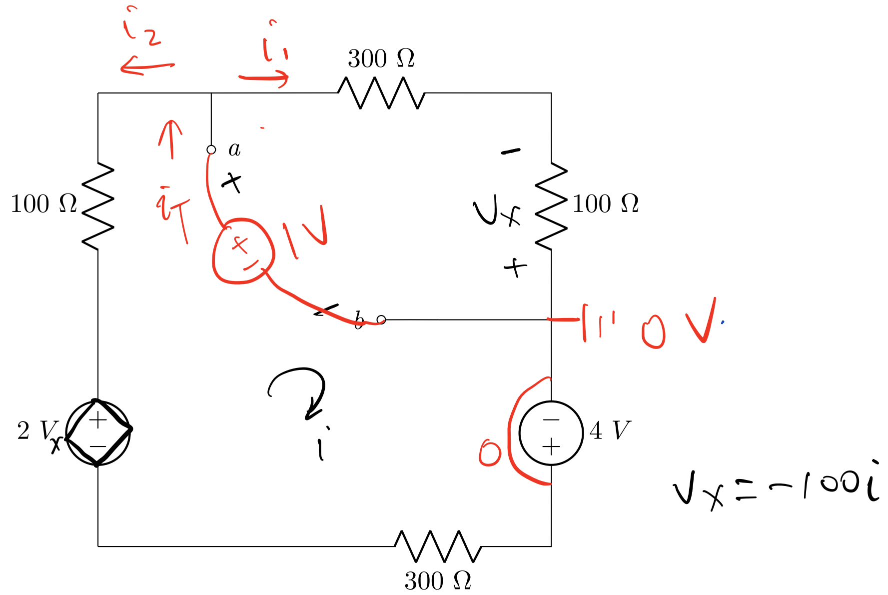
$= 8.12 \times 10^4 \times 50 \times 10^{-6}$

$= 4.06V$

Total Section 1: [16]

Section 2

Questions 13 to 14 [4]		
Determine the Thevenin equivalent voltage V_{Th} and the Thevenin equivalent resistance R_{Th} with respect to terminals $a - b$.		
13	V_{Th}. [2]	1.6 V
14	R_{Th}. [2]	160 Ω



(13)

$$-2V_x + 100i + 300i + 100i - 4 + 300i = 0$$

$$200i + 800i = 4 \Rightarrow i = \frac{4}{1000}\ A$$

$$V_{Th} = 400i = 1.6\ V.$$

(14) Add 1 V, switch 4V to short circuit.

$$i_T = i_1 + i_2, \quad i_1 = \frac{1}{400}, \quad V_x = -\frac{1}{4}\ V.$$

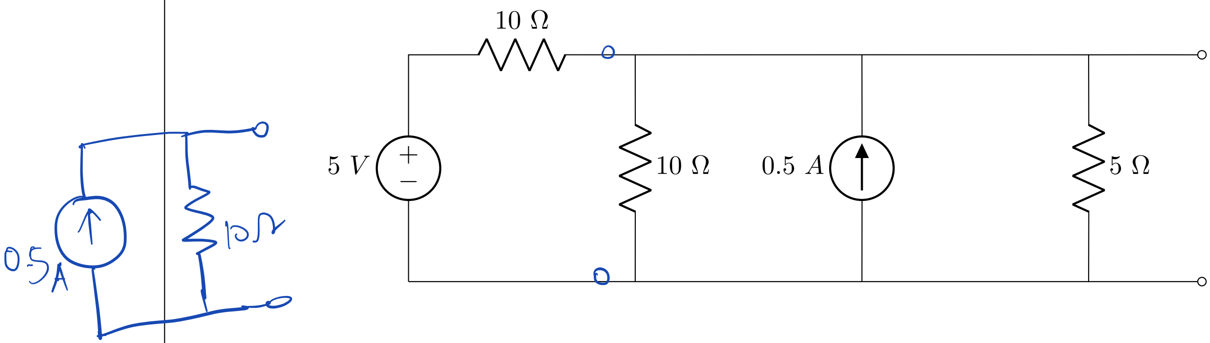
$$i_2 = \frac{1 - 2V_x}{400} = \frac{1.5}{400}, \quad i_T = \frac{1 + 1.5}{400}$$

$$R_{Th} = \frac{400}{2.5} = 160\ \Omega.$$

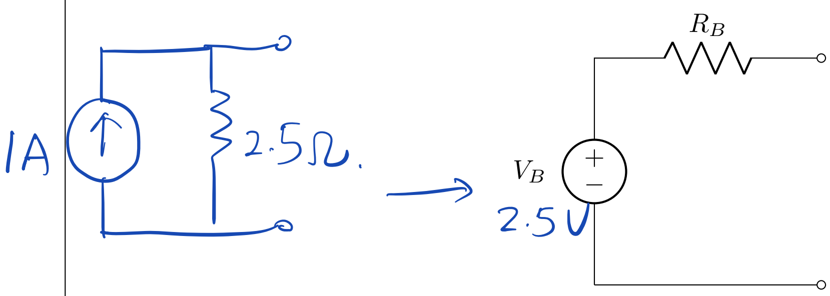
Question 15 [2]

Use source transformation to obtain Circuit B so that it is equivalent to Circuit A.

Circuit A:



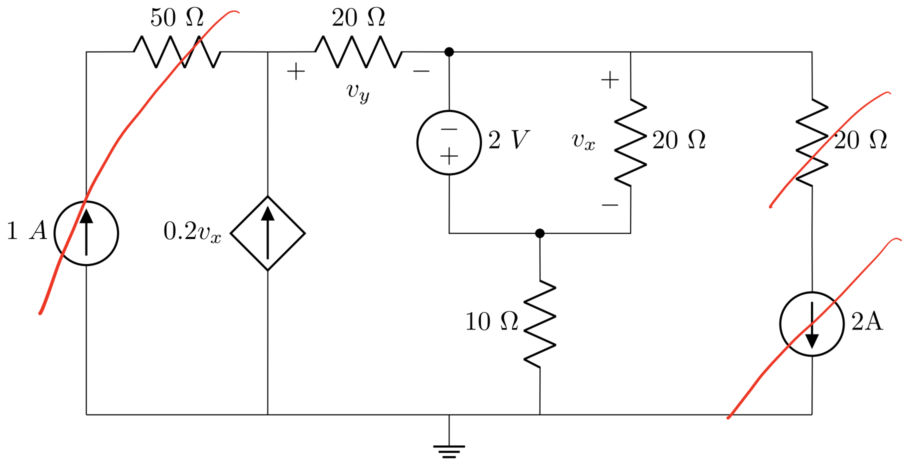
Circuit B:



15	Calculate the value of the voltage source V_B. [2] 2.5V

Question 16 [2]

Determine v_y^{2V}, the independent contribution of the 2 V voltage source to v_y.



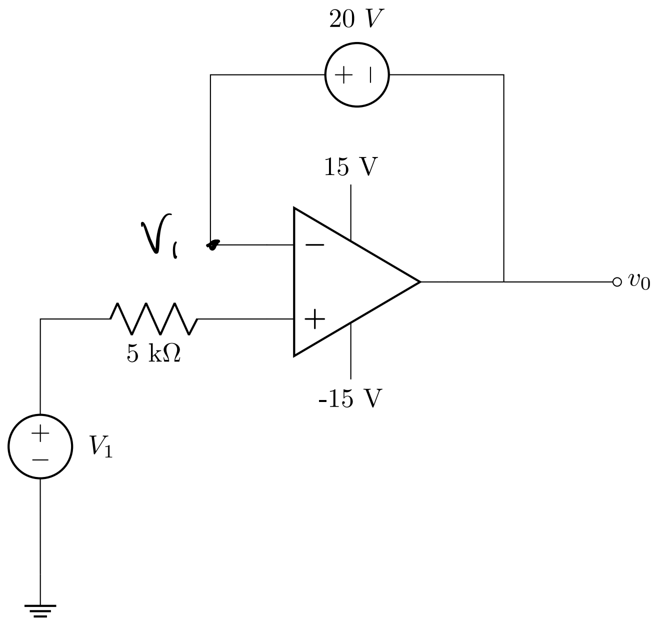
16	$v_y^{2\ V}$ [2]	$-8V$

$V_x = -2V.$

$$V_y = 20 \times 0.2 V_x$$
$$= 20 \times 0.2 \times (-2)$$
$$= -8V.$$

Question 17 [2]

Assume the op-amp is ideal. Determine the minimum value of V_1 before the op-amp goes into negative saturation.



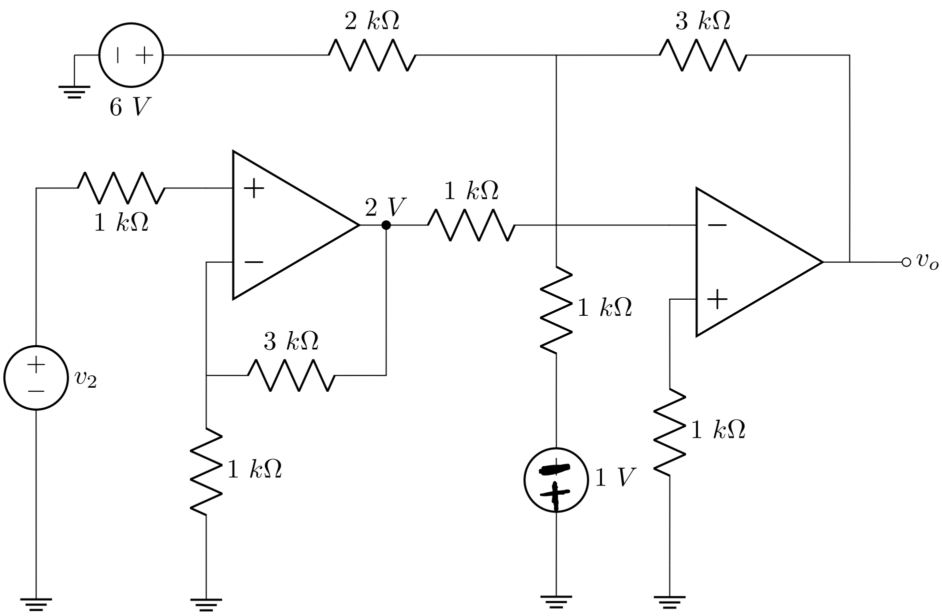
17	V_1 [2]	5 V

$V_0 = V_1 - 20$

$-15 \leq V_1 - 20 \leq 15$

$5 \leq V_1 \leq 35$

Questions 18 to 19 [4]	
Assume that the op-amps are ideal.	



18	Calculate the value of the voltage source v_2. [2] 0.5V
19	Calculate the output voltage v_o. [2] -12V

(18) $2 = V_2\left(1+\frac{3}{1}\right) \Rightarrow V_2 = 0.5V$

(19) $V_0 = -\left(\frac{3}{2}\cdot 6 + \frac{3}{1}\cdot 2 + \frac{3}{1}(-1)\right)$

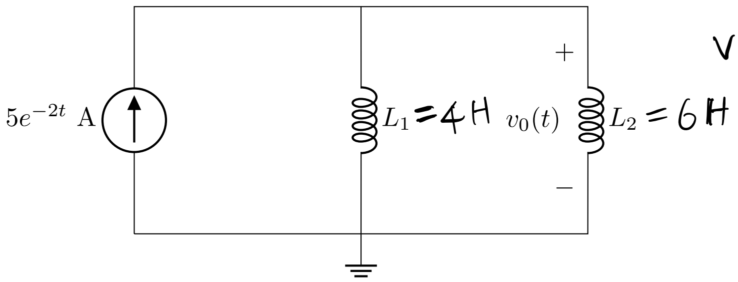
$= -12V.$

Total Section 2: [14]

Section 3

Question 20 [2]

$v(t) = L\frac{di}{dt}$



$L = 4 \,||\, 6 = 2.4H.$

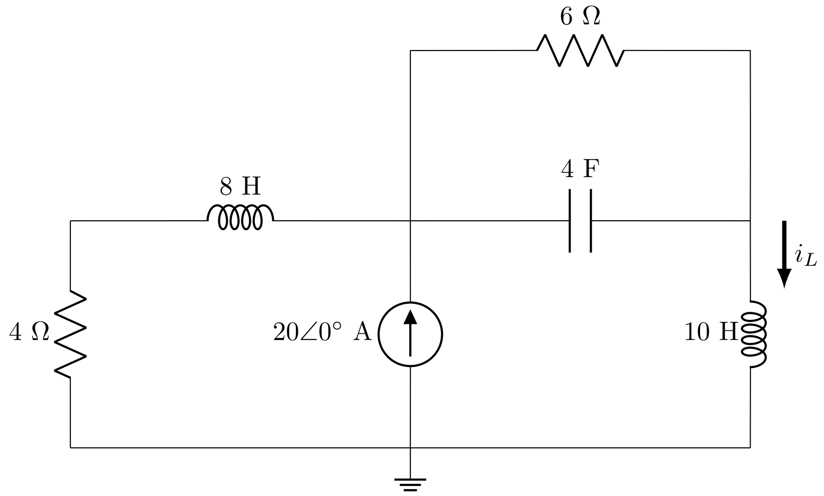
20	Calculate $v_0(2)$, the voltage $v_0(t)$ at $t = 2$ second. [2] -0.4396

$V_o(t) = 2.4\,(5e^{-2t})'$

$= 2.4 \cdot 5e^{-2t} \cdot (-2)$

$= -24e^{-2t}$

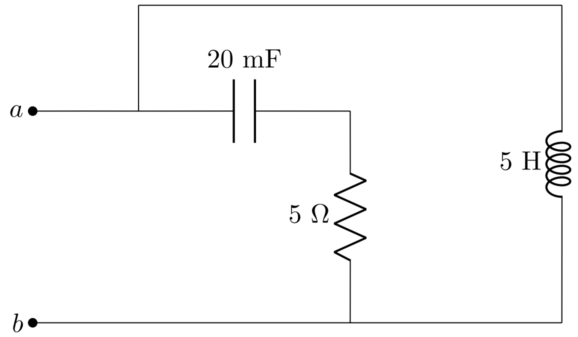
$V_o(2) = -24e^{-4} = -0.4396$

Question 21 [2]	
The circuit below is under steady DC conditions.	
	
21	Find the **energy** w_C stored in the capacitor. [2] 4608 J

$i_L = 20 \cdot \frac{4}{4+6} = 8A$, $V_C = 6 \times 8 = 48V$.

$w_C = \frac{1}{2} C V_C^2 = \frac{1}{2} \times 4 \times 48^2 = 4608 J$

Question 22 [2]



$2+j16\ \Omega$

22	Calculate $\mathbf{Z}_{ab}$, the equivalent impedance with respect to terminals $a-b$ at $\omega = 2$ rad/s. [2]

$$Z_C = \frac{1}{j2 \times 20 \times 10^{-3}} = -j25\ \Omega.$$

$$Z_C = j\omega L = j \times 5 \times 2 = j10$$

$$Z_{ab} = (5-j25) \,||\, j10 = \frac{j10(5-j25)}{5-j15}$$

$$= 2+j16\ \Omega.$$

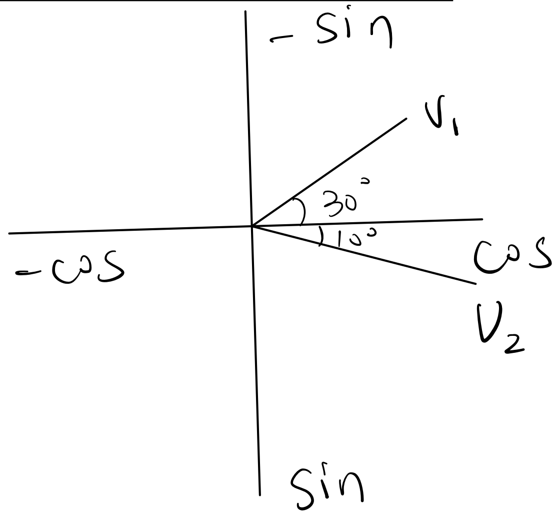
Questions 23 to 25 [4]	
Consider the following two sinusoid waveforms: $v_1(t) = -30\sin(\omega t - 60^\circ)$ and $v_2(t) = 40\cos(\omega t - 10^\circ)$.	
23	Choose the correct statement. 1. v_1 leads v_2. 2. v_2 leads v_1. Write 1 if **statement 1** is correct **or** 2 if **statement 2** is correct. [1]
24	Determine the phase angle ϕ_d between the two sinusoids, with $\phi_d \in [0^\circ, 180^\circ]$. [2]
25	Use phasors to find $\mathbf{V} = \mathbf{V}_1 + \mathbf{V}_2$. [1]

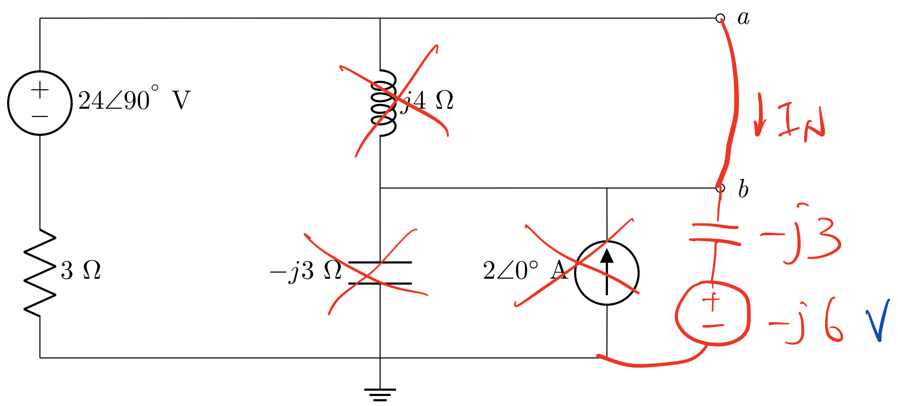
$\phi_d = 40^\circ$

$V_1 = 30\angle 30^\circ$

$V_2 = 40\angle -10^\circ$

$V = 65.87\angle 7.024^\circ$



Questions 26 to 27 [4]		
Find the Norton equivalent current $\mathbf{I}_N$ and the Norton equivalent impedance $\mathbf{Z}_N$ at terminals *a*-*b*.		
		
26	$\mathbf{I}_N$ [2]	$-3+j3$ A
27	$\mathbf{Z}_N$ [2]	$4.8+j2.4\ \Omega$

26

Use source transformation:

$$3I_N - j24 - j3I_N + j6 = 0$$

$$(3-j3)I_N = j18$$

$$I_N = \frac{j18}{3-j3}$$

$$= -3+j3\,A$$

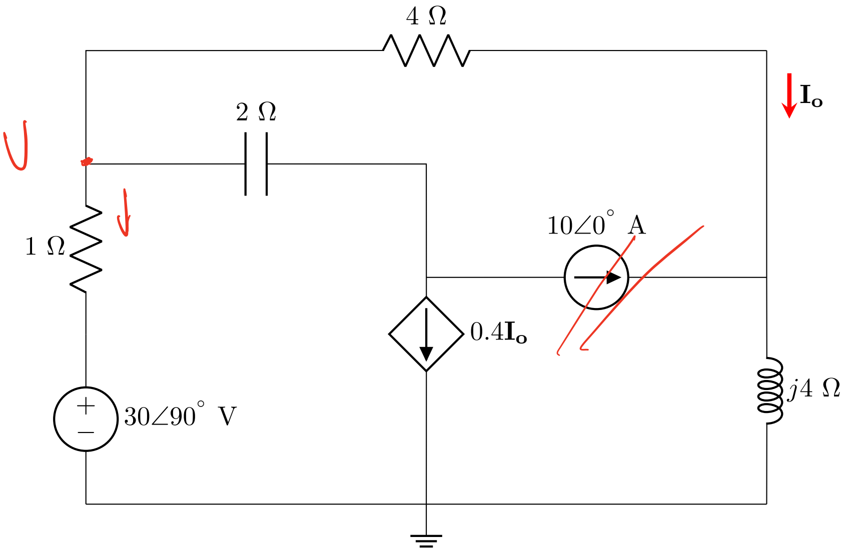
$$Z_N = j4 \,||\, (3-j3)$$

$$= \frac{j4(3-j3)}{3+j}$$

$$= 4.8+j2.4\,\Omega$$

Question 28 [2]

Determine $\mathbf{I_o^V}$, the exclusive contribution of the **voltage** source to $\mathbf{I}_o$.



28	$\mathbf{I_o^V}$ [2]	4.464∠53.47° A

$$I_o = \frac{V}{4+j4}$$

$$\frac{V - j30}{1} + 0.4I_o + I_o = 0$$

$$V - j30 = \frac{-1.4V}{4+j4}$$

$$V + \frac{1.4}{4+j4}V = j30$$

$$\frac{5.4+j4}{4+j4}V = j30$$

$$V = \frac{j30(4+j4)}{5.4+j4}$$

$$I_o^V = \frac{V}{4+j4}$$

$$= \frac{j30}{5.4+j4}$$

$$= 4.464\angle 53.47^\circ A$$

Total Section 3: [16]

Grand Total : [46]